

Quantum Mechanics 3 – Question 2, Task 7 Ext.

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2. The Particle in a One-Dimensional Infinite Potential Well – Including Task 7 Extension

The infinite potential well (or “particle in a box”) is one of the simplest exactly solvable quantum systems. Although idealised, it demonstrates several fundamental concepts of quantum mechanics, including wavefunctions, quantised energy levels and Heisenberg’s Uncertainty Principle.

$$\psi(x, t)$$

describes the wavefunction of a particle of mass m in a ‘box’ of potential energy V .

The potential is defined as

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & \text{otherwise.} \end{cases}$$

2(i) Standing Waves

The particle is confined to the region $0 \leq x \leq a$ by an infinite potential outside the box. Therefore, the wavefunction must satisfy the boundary conditions

$$\psi(0, t) = \psi(a, t) = 0$$

At the left-hand boundary, $x = 0$, the sine term becomes

$$\sin(0) = 0,$$

so that $\psi(0, t) = 0$. At the right-hand boundary, $x = a$,

$$\sin\left(\frac{n\pi a}{a}\right) = \sin(n\pi) = 0$$

This means the points $x = 0$ and $x = a$ are nodes of a standing wave.

The condition above is satisfied when n is an integer. The value $n = 0$ would result in $\psi(x, t) = 0$ everywhere and therefore cannot represent a particle, so the allowed values are

$$n = 1, 2, 3, \dots$$

Thus, n is a positive integer and represents the quantum number associated with the standing-wave state.

$$\cos(\omega t)$$

where $\omega = 2\pi f$, $f = c/\lambda$, describes the wavefunction's sinusoidal time dependence. At a fixed position x , the wavefunction therefore oscillates with angular frequency ω :

$$\psi(x, t) \propto \cos(\omega t)$$

The angular frequency is related to the energy of the state by

$$E = \hbar\omega$$

A suitable form for the wavefunction is therefore

$$\psi(x, t) = A_n \sin\left(\frac{n\pi x}{a}\right) \cos(\omega t)$$

A_n is the normalisation factor. The sine term describes the spatial standing-wave pattern, while the cosine term describes its time dependence.

2(ii) A Solution to the Schrödinger Equation

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = E\psi$$

Inside the box, the potential is zero, and the time-independent Schrödinger equation is therefore

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi$$

Consider the proposed wavefunction

$$\psi(x, t) = A_n \sin\left(\frac{n\pi x}{a}\right) \cos(\omega t)$$

Differentiating twice with respect to x gives

$$\frac{\partial^2 \psi}{\partial x^2} = -A_n \left(\frac{n\pi}{a}\right)^2 \sin\left(\frac{n\pi x}{a}\right) \cos(\omega t) = -\left(\frac{n\pi}{a}\right)^2 \psi$$

Substituting this into the Schrödinger equation gives

$$-\frac{\hbar^2}{2m} \left[-\left(\frac{n\pi}{a}\right)^2 \psi \right] = E\psi$$

Hence,

$$E(n, a) = \frac{n^2 \pi^2 \hbar^2}{2ma^2}$$

For a proton confined to a box of width

$$a = 0.47 \times 10^{-15} \text{ m},$$

the ground-state energy ($n = 1$) is approximately

$$E_1 \approx 927 \text{ MeV}$$

Since the energy varies with n^2 , the general energy is

$$E_n \approx 927n^2 \text{ MeV}$$

The proton rest mass energy is

$$m_p c^2 \approx 938 \text{ MeV}$$

Consequently,

$$\frac{E_1}{m_p c^2} \approx \frac{927}{938} \approx 0.988$$

The ground-state energy is approximately 98.8% of the proton's rest mass energy, $m_p c^2$. The non-relativistic Schrödinger equation assumes that the kinetic energy is much smaller than the rest energy, so that

$$K_e \approx \frac{p^2}{2m}.$$

However, in this case, the predicted confinement energy is comparable to $m_p c^2$. This indicates that the proton would have a sufficiently large momentum for relativistic effects to become important. The non-relativistic particle-in-a-box model no longer provides an accurate description of the proton in such a small confinement region. A relativistic approach would instead be required.

2(iii) Time Dependence and Normalisation

The time-dependent factor $\cos(\omega t)$ can be replaced by the complex exponential $e^{-iEt/\hbar}$. This follows from the relationship between energy and angular frequency,

$$E = \hbar\omega$$

Thus, the wavefunction can be written as

$$\psi_n(x, t) = A_n \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}$$

The Born interpretation states that $|\psi(x, t)|^2$ represents the probability density of finding the particle at position x . Since the particle must be found somewhere within the box, the total probability must be total certainty:

$$\int_0^a |\psi_n(x, t)|^2 dx = 1$$

Taking the modulus squared of the wavefunction gives

$$|\psi_n(x, t)|^2 = A_n^2 \sin^2\left(\frac{n\pi x}{a}\right) \left|e^{iE_n t/\hbar}\right|^2$$

Since

$$\left|e^{-iE_n t/\hbar}\right|^2 = 1,$$

the probability density is independent of time:

$$|\psi_n(x, t)|^2 = A_n^2 \sin^2\left(\frac{n\pi x}{a}\right)$$

Normalisation therefore requires

$$A_n^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$$

Using

$$\int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2},$$

we obtain

$$A_n^2 \frac{a}{2} = 1$$

Therefore,

$$\boxed{A_n = \sqrt{\frac{2}{a}}}$$

The normalised wavefunction is

$$\boxed{\psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}}$$

2(iv) Expected Values of Momentum

The momentum operator in one dimension is

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}$$

The expected value of momentum is given by

$$\langle p \rangle = \int_0^a \psi^* \hat{p} \psi dx$$

For the normalised wavefunction

$$\psi_n(x, t) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar},$$

differentiation with respect to x gives

$$\frac{\partial \psi_n}{\partial x} = \sqrt{\frac{2}{a}} \frac{n\pi}{a} \cos\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar}$$

Therefore,

$$\langle p \rangle = -i\hbar \frac{2n\pi}{a^2} \int_0^a \sin\left(\frac{n\pi x}{a}\right) \cos\left(\frac{n\pi x}{a}\right) dx$$

The integral evaluates to zero, giving

$$\boxed{\langle p \rangle = 0}$$

Physically, this occurs because the standing wave contains equal components travelling in opposite directions, corresponding to equal and opposite momenta.

The square of the momentum operator is

$$\hat{p}^2 = -\hbar^2 \frac{\partial^2}{\partial x^2}$$

From the second derivative of the wavefunction,

$$\frac{\partial^2 \psi_n}{\partial x^2} = -\left(\frac{n\pi}{a}\right)^2 \psi_n$$

Substituting,

$$\hat{p}^2 \psi_n = \frac{n^2 \pi^2 \hbar^2}{a^2} \psi_n$$

Taking the expected value gives

$$\langle p^2 \rangle = \frac{n^2 \pi^2 \hbar^2}{a^2} \int_0^a |\psi_n|^2 dx$$

Since the wavefunction is normalised,

$$\int_0^a |\psi_n|^2 dx = 1,$$

and hence

$$\boxed{\langle p^2 \rangle = \frac{n^2 \pi^2 \hbar^2}{a^2}}$$

2(v)(a) Expected Values of x

The expected value of x is given by

$$\langle x \rangle = \int_0^a x |\psi_n|^2 dx = \frac{2}{a} \int_0^a x \sin^2 \left(\frac{n\pi x}{a} \right) dx$$

Quoting the integral identity:

$$\int x \sin^2(\alpha x) dx = \frac{1}{4}x^2 - \frac{x \sin(2\alpha x)}{4\alpha} - \frac{\cos(2\alpha x)}{8\alpha^2}$$

Let

$$\alpha = \frac{n\pi}{a}$$

Evaluating the definite integral from $x = 0$ to $x = a$ (noting that $2\alpha a = 2n\pi$, so $\sin(2n\pi) = 0$ and $\cos(2n\pi) = 1$):

$$\int_0^a x \sin^2(\alpha x) dx = \left[\frac{1}{4}x^2 - \frac{x \sin(2\alpha x)}{4\alpha} - \frac{\cos(2\alpha x)}{8\alpha^2} \right]_0^a \quad (1)$$

$$= \left(\frac{1}{4}a^2 - 0 - \frac{1}{8\alpha^2} \right) - \left(0 - 0 - \frac{1}{8\alpha^2} \right) \quad (2)$$

$$= \frac{1}{4}a^2 \quad (3)$$

Thus:

$$\boxed{\langle x \rangle = \frac{2}{a} \left(\frac{1}{4}a^2 \right) = \frac{1}{2}a}$$

2(v)(b) Calculation of $\langle x^2 \rangle$

The expected value $\langle x^2 \rangle$ is:

$$\langle x^2 \rangle = \int_0^a x^2 |\psi_n(x)|^2 dx = \frac{2}{a} \int_0^a x^2 \sin^2(\alpha x) dx$$

Quoting the integral identity:

$$\int x^2 \sin^2(\alpha x) dx = \frac{1}{6}x^3 - \frac{x \cos(2\alpha x)}{4\alpha^2} - \frac{(2\alpha^2 x^2 - 1) \sin(2\alpha x)}{8\alpha^3}$$

Let

$$\alpha = \frac{n\pi}{a}$$

Evaluating from $x = 0$ to $x = a$:

$$\begin{aligned} \int_0^a x^2 \sin^2(\alpha x) dx &= \left[\frac{1}{6}x^3 - \frac{x \cos(2\alpha x)}{4\alpha^2} - \frac{(2\alpha^2 x^2 - 1) \sin(2\alpha x)}{8\alpha^3} \right]_0^a \\ &= \left(\frac{1}{6}a^3 - \frac{a}{4\alpha^2} - 0 \right) - (0 - 0 - 0) \\ &= \frac{a^3}{6} - \frac{a}{4\alpha^2} \end{aligned}$$

Substituting $\alpha = \frac{n\pi}{a}$:

$$\int_0^a x^2 \sin^2(\alpha x) dx = \frac{a^3}{6} - \frac{a^3}{4n^2\pi^2}$$

And

$$\begin{aligned} \langle x^2 \rangle &= \frac{2}{a} \left(\frac{a^3}{6} - \frac{a^3}{4n^2\pi^2} \right) \\ &= \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} \\ &= \frac{1}{3}a^2 \left(1 - \frac{3}{2n^2\pi^2} \right) \end{aligned}$$

2(vi)(a) Calculation of Δx

The uncertainty in position Δx is defined by:

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$$

Using the previously derived results of $\langle x \rangle$ and $\langle x^2 \rangle$:

$$\begin{aligned} (\Delta x)^2 &= \left(\frac{a^2}{3} - \frac{a^2}{2n^2\pi^2} \right) - \left(\frac{a}{2} \right)^2 \\ &= \left(\frac{a^2}{3} - \frac{a^2}{4} \right) - \frac{a^2}{2n^2\pi^2} \\ &= \frac{a^2}{12} - \frac{a^2}{2n^2\pi^2} \end{aligned}$$

Factoring out $\frac{a^2}{4n^2\pi^2}$:

$$\begin{aligned}(\Delta x)^2 &= \frac{a^2}{4n^2\pi^2} \left(\frac{4n^2\pi^2}{12} - 2 \right) \\ &= \frac{a^2}{4n^2\pi^2} \left(\frac{n^2\pi^2}{3} - 2 \right)\end{aligned}$$

Taking the square root yields

$$\Delta x = \frac{a}{2n\pi} \sqrt{\frac{n^2\pi^2}{3} - 2}$$

2(vi)(b) Calculation of Δp

The uncertainty in momentum Δp is defined by:

$$(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2$$

Using the previously derived results of $\langle p \rangle$ and $\langle p^2 \rangle$:

$$\Delta p = \sqrt{\frac{n^2\pi^2\hbar^2}{a^2} - 0} = \frac{n\pi\hbar}{a}$$

2(vi)(c) Product of Uncertainties $\Delta x \Delta p$

Multiplying Δx and Δp :

$$\begin{aligned}\Delta x \Delta p &= \left(\frac{a}{2n\pi} \sqrt{\frac{n^2\pi^2}{3} - 2} \right) \left(\frac{n\pi\hbar}{a} \right) \\ &= \frac{1}{2} \hbar \sqrt{\frac{n^2\pi^2}{3} - 2}\end{aligned}$$

2(vi)(d) Verification of the Uncertainty Principle

Heisenberg's Uncertainty Principle requires that:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

For our result to satisfy this inequality, we need:

$$\sqrt{\frac{n^2\pi^2}{3} - 2} \geq 1$$

Testing for the ground state ($n = 1$):

$$\frac{(1)^2\pi^2}{3} - 2 \approx 1.2899 > 1$$

Taking the square root:

$$\sqrt{1.2899} \approx 1.1357 > 1$$

Since $\sqrt{\frac{n^2\pi^2}{3}} - 2$ increases as n increases, the factor is strictly greater than 1 for all quantum numbers n , therefore,

$$\Delta x \Delta p > \frac{\hbar}{2} \quad \text{for all } n = 1, 2, 3, \dots$$

This verifies that the particle in a box always satisfies Heisenberg's Uncertainty Principle.